

Project Design Phase-II Data Flow Diagram & User Stories

Date	01-10-2026
Team ID	SWTID-2026-8759
Project Name	Import Data using Transform Maps
Maximum Marks	4 Marks

Data Flow Diagrams:

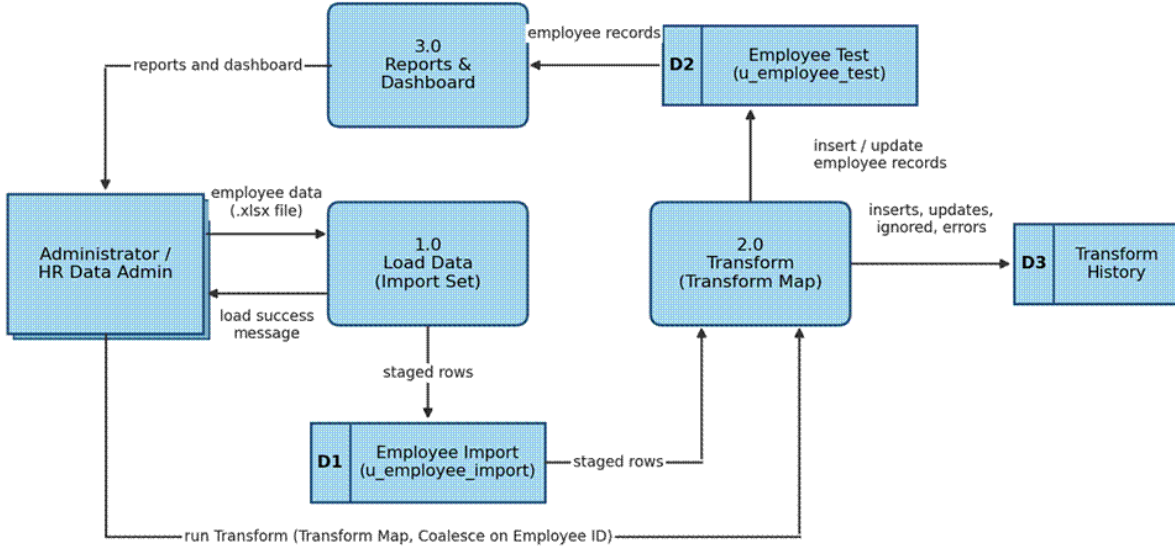
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Flow of data in this project:

1. Administrator prepares the employee spreadsheet and downloads it as Sample Spreadsheet.xlsx.
2. Administrator uploads the file with Load Data; ServiceNow stages the rows in the Employee Import table (u_employee_import).
3. Administrator creates a Transform Map from Employee Import to Employee Test and maps the five fields; Coalesce is enabled on Employee ID.
4. Running Transform inserts new records and updates existing ones in Employee Test (u_employee_test); the counts are logged in Transform History.
5. Reports (by Department, by Location, and a list) are built on Employee Test and shown on the Employee Analytics Dashboard.

Example: DFD Level 0 for Import Data using Transform Maps

DFD Level 0: Import Data using Transform Maps



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Data Preparation	USN-1	As an administrator, I can create a spreadsheet with employee ID, name, email, department and location and download it as an .xlsx file.	The file has 15 records and 5 columns and opens in Excel.	High	Sprint-1
	Custom Table	USN-2	As an administrator, I can create the custom table Employee Test (u_employee_test) with five string fields.	The table and all five fields appear on the Employee Test form.	High	Sprint-1
	Data Loading (Import Set)	USN-3	As an administrator, I can upload the .xlsx file using Load Data to create the Employee Import staging table.	Progress shows State Complete and Completion code Success with 15 rows processed.	High	Sprint-1
	Transform Map	USN-4	As an administrator, I can create a Transform Map from Employee Import to Employee Test and map the fields.	The Field Maps list shows 5 mapped fields.	High	Sprint-1

		USN-5	As an administrator, I can run the Transform to move staged records into the target table.	Transformation shows Complete and Success; 15 records appear in Employee Tests.	High	Sprint-1
	Duplicate Prevention (Coalesce)	USN-6	As an administrator, I can enable Coalesce on Employee ID so that re-imports update existing records.	Re-importing changed rows updates them; no Employee ID is duplicated.	High	Sprint-2
		USN-7	As an administrator, I can re-import the same file without creating duplicates.	Transform History shows 0 inserts, 0 updates and 4 ignored rows.	Medium	Sprint-2
		USN-8	As an administrator, I can review Transform History to see inserted, updated, ignored and error counts.	Transform History lists total, inserts, updates, ignored, skipped and errors for each run.	Medium	Sprint-2
	Automation	USN-9	As an administrator, I can schedule the import through a data source so that new files load automatically.	A scheduled import runs without manual upload.	Low	Sprint-4
HR Manager	Reports	USN-10	As an HR manager, I can view employee count by department in a pie chart.	The chart shows ServiceNow 11, Salesforce 4 and Aiml 2.	Medium	Sprint-3
		USN-11	As an HR manager, I can view employee count by location in a bar chart.	The chart shows counts for Hyderabad and Chennai.	Medium	Sprint-3
		USN-12	As an HR manager, I can view a list report of all employee records.	The list shows Employee ID, Name, Email, Department and Location for all 17 records.	Medium	Sprint-3
	Dashboard	USN-13	As an HR manager, I can see all three reports together on the Employee Analytics Dashboard.	The dashboard shows the pie, bar and list widgets.	High	Sprint-3

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Source Data Preparation	Create employee data in a spreadsheet (Employee ID, Name, Email, Department, Location) Export the spreadsheet as an Excel (.xlsx) file
FR-2	Target Table Creation	Create custom table Employee Test (u_employee_test) Add String fields: Employee ID, Employee Name, Email, Department, Location
FR-3	Data Loading (Import Set)	Upload the .xlsx file through Load Data into the staging table Employee Import (u_employee_import)

		Load further files into the existing import set table
FR-4	Field Mapping (Transform Map)	Create Transform Map Sample Spreadsheet Import from source table to target table
		Map source fields to target fields using Auto Map Matching Fields / Mapping Assist
FR-5	Transformation & Validation	Run the transform and confirm State = Complete, Completion code = Success
		Verify records in the Employee Tests table and review Transform History (inserts, updates, ignored, errors)
FR-6	Duplicate Prevention (Coalesce)	Enable Coalesce on Employee ID so repeated imports update existing records instead of inserting duplicates
FR-7	Reporting	Create reports on the Employee Test table: Employees by Department (pie), Employees by Location (bar), Employee List Report (list)
FR-8	Dashboard & Access	Create the Employee Analytics Dashboard and add all three reports
		Restrict dashboard access by users, groups and roles through the Share option

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Guided, low-code steps (Load Data, Auto Map, Transform) that beginners can follow without scripting. Dashboard presents data through simple charts and a list view
NFR-2	Security	Dashboard sharing restricted by users, groups and roles. Dashboard creation controlled through the pa_dashboards ACL. Imports run by authorized admin users
NFR-3	Reliability	Coalesce on Employee ID keeps repeated imports consistent. A re-import of identical data gives 0 inserts, 0 updates, 4 ignored. Import and transform logs show errors
NFR-4	Performance	The 15-record import and the 4-record re-import each complete within seconds (about 1 second in the transform history)
NFR-5	Availability	Runs on the cloud-hosted ServiceNow instance, so import data, reports and the dashboard are available whenever the instance is up
NFR-6	Scalability	The same Import Set and Transform Map approach handles larger spreadsheets and repeated loads. More reports and widgets can be added to the dashboard

Project Design Phase-II

Technology Stack (Architecture & Stack)

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Technical Architecture

Example flow: employee data prepared in a spreadsheet is staged in an Import Set table, transformed into a custom target table by a Transform Map, then reported on in a dashboard.

1. User prepares employee data in Google Sheets and downloads it as .xlsx (local / external)
2. User uploads the file through Load Data into the staging table u_employee_import (ServiceNow cloud)
3. Transform Map "Sample Spreadsheet Import" maps source fields to the target table u_employee_test, with Coalesce on Employee ID (ServiceNow cloud)
4. Transformed records are stored in the Employee Test table (ServiceNow cloud database)
5. Reports (pie, bar, list) read from Employee Test and are shown on the Employee Analytics Dashboard (ServiceNow cloud)

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web UI used to load data, configure maps, and view reports and the dashboard	ServiceNow Now Platform UI (Application Navigator, forms, lists) in a web browser

2	Application Logic-1	Loads the .xlsx file into a staging (Import Set) table	ServiceNow Import Sets / Load Data
3	Application Logic-2	Maps source fields to target fields and creates or updates records; Coalesce on Employee ID prevents duplicates	ServiceNow Transform Map (Auto Map Matching Fields, Mapping Assist, Coalesce)
4	Application Logic-3	Builds reports and dashboards on employee data	ServiceNow Reports and Platform Analytics Dashboards
5	Database	Staging table u_employee_import and target table u_employee_test (Employee ID, Employee Name, Email, Department, Location, all String)	ServiceNow relational database (custom tables)
6	Cloud Database	Database service on the cloud instance that stores all records	ServiceNow cloud instance database
7	File Storage	Source file Sample Spreadsheet.xlsx uploaded during Load Data	Local file system (download from Google Sheets) and ServiceNow attachment storage
8	External API-1	Not used. Google Sheets is used manually to prepare and export the .xlsx file	N/A
9	External API-2	Not used	N/A
10	Machine Learning Model	Not used	N/A

11	Infrastructure (Server / Cloud)	Application is configured and run on a ServiceNow cloud instance. Local Server Configuration: none, a web browser is the only client. Cloud Server Configuration: ServiceNow-hosted instance, Global application scope	Cloud (ServiceNow Now Platform)
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Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source frameworks. The solution is built with ServiceNow's native low-code features	ServiceNow Now Platform (Import Sets, Transform Maps, Reports)
2	Security Implementations	Role-based access, ACL pa_dashboards (create operation with Admin overrides) controls dashboard creation, dashboard sharing restricted by users, groups and roles	ServiceNow ACLs, roles, dashboard Share settings
3	Scalable Architecture	Staging table separates loading from transformation, so larger files and repeated loads reuse the same Transform Map. More reports and widgets can be added to the dashboard	ServiceNow multi-tier platform architecture (Import Set staging, Transform Map, target table)

4	Availability	Instance is hosted and managed by ServiceNow in the cloud, so the application is reachable from any browser	ServiceNow cloud hosting
5	Performance	Batch transform of a staged import set (about 1 second for the sample loads), Coalesce on Employee ID avoids duplicate checks on every field, reports aggregate directly on the target table	ServiceNow Transform Engine, Coalesce, Platform Analytics reports